

PROJECT DESIGN PHASE

Proposed Solution Template

Date	15 February 2026
Team ID	T-12
Project Name	FinVault – Virtual Stock Trading Platform
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Real-world stock market trading carries significant financial risk, preventing students and beginners from learning essential trading concepts through experience. Traditional platforms are either overly complex for educational environments or lack structured dashboard tracking and user customizability. FinVault solves this by providing a safe, sandbox virtual trading desktop with live-updating stock prices and profile configurations.
2.	Idea / Solution description	FinVault is a MERN stack virtual stock simulator. Regular users sign up, customize their profile with an avatar (handled via Multer middleware), and receive a virtual wallet loaded with \$100,000 mock cash. They can search listed stocks, watch assets on a watchlist, analyze historical trends via Chart.js line plots, and submit simulated BUY/SELL market orders. Admin users leverage a monitoring dashboard to manage users, add/edit/delete stocks, and audit the global ledger.
3.	Novelty / Uniqueness	• Stateless JWT Route Protection: Employs robust industry-standard JSON Web Tokens to secure APIs and enforce role-based routing (Admin vs. User). • Multer Profile Integration: Connects profile avatar file-uploads directly to database schemas. • Atomic Operations: Trade execution processes (wallet updates and portfolio adjustments) occur inside unified atomic operations to prevent data discrepancies.
4.	Social Impact / Customer Satisfaction	• Financial Literacy: Lowers the learning curve and entry barriers to financial markets, educating high school and university students risk-free. • User Interface (UX): Delivers clean, responsive dark-themed designs with interactive stock tooltips, real-time balance calculations, and robust transaction logs to track learning outcomes.
5.	Business Model (Revenue Model)	• B2B Institutional Licensing: White-label the platform for schools, universities, and brokerage firms to host private trading competitions. • Freemium Subscriptions: Free core sandbox access for individual learners, with premium tiers for advanced portfolio analytics and data report exports. • Affiliate Referral: Future integration of referral links to real brokerages once users demonstrate trading competency.
6.	Scalability of the Solution	Decoupled architecture supports high horizontal scalability. The frontend React client can be statically hosted on global CDN edges (like Vercel) for sub-second page loads. The Express backend API gateway is stateless, allowing deployment on auto-scaling container configurations (Render/AWS ECS). MongoDB Atlas handles high-frequency writes via replica sets and database sharding.